

TAILOR: An Effective Selection Framework for Enhancing Synthetic Tabular Data Quality via LLM-Guided MCTS (Technical Appendix)

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A Full Ablation Results

Table 6 completes the ablation of Table 3 on the remaining three datasets.

Table 6: Ablation results on the remaining datasets. Format follows Table 3.

Data	Method	Shape	Trend	DCR	Util.	MOS	Imp.
Magic	Raw	.956 $\pm$.000	.966 $\pm$.002	.926 $\pm$.001	.932 $\pm$.002	.823 $\pm$.001	0.00%
	Select	.925 $\pm$.014	.944 $\pm$.009	.700 $\pm$.029	.924 $\pm$.003	.875 $\pm$.006	+6.35%
	+Floor	.967 $\pm$.002	.958 $\pm$.002	.877 $\pm$.002	.923 $\pm$.002	.836 $\pm$.001	+1.52%
	+Repair	.951 $\pm$.007	.955 $\pm$.003	.608 $\pm$.045	.932 $\pm$.004	.913 $\pm$.011	+10.97%
	Greedy	.946 $\pm$.011	.948 $\pm$.009	.584 $\pm$.055	.936 $\pm$.004	.920 $\pm$.011	+11.73%
	w/o LLM	.948 $\pm$.011	.951 $\pm$.009	.615 $\pm$.055	.934 $\pm$.004	.917 $\pm$.011	+11.39%
	TAILOR	.943 $\pm$.011	.942 $\pm$.009	.540 $\pm$.055	.939 $\pm$.003	.925 $\pm$.011	+12.36%
News	Raw	.963 $\pm$.000	.981 $\pm$.002	.936 $\pm$.001	—	.728 $\pm$.005	0.00%
	Select	.952 $\pm$.002	.974 $\pm$.003	.728 $\pm$.034	—	.784 $\pm$.010	+7.65%
	+Floor	.967 $\pm$.001	.977 $\pm$.002	.912 $\pm$.001	—	.735 $\pm$.005	+0.96%
	+Repair	.960 $\pm$.003	.976 $\pm$.003	.625 $\pm$.059	—	.800 $\pm$.013	+9.94%
	Greedy	.956 $\pm$.004	.974 $\pm$.004	.612 $\pm$.059	—	.805 $\pm$.014	+10.62%
	w/o LLM	.958 $\pm$.004	.975 $\pm$.004	.638 $\pm$.059	—	.803 $\pm$.014	+10.31%
	TAILOR	.953 $\pm$.004	.973 $\pm$.004	.578 $\pm$.059	—	.809 $\pm$.014	+11.17%
Shoppers	Raw	.945 $\pm$.001	.952 $\pm$.001	.927 $\pm$.001	.908 $\pm$.003	.806 $\pm$.002	0.00%
	Select	.939 $\pm$.004	.947 $\pm$.003	.665 $\pm$.031	.903 $\pm$.004	.878 $\pm$.009	+8.90%
	+Floor	.962 $\pm$.001	.962 $\pm$.001	.864 $\pm$.001	.910 $\pm$.003	.833 $\pm$.002	+3.32%
	+Repair	.952 $\pm$.003	.952 $\pm$.004	.610 $\pm$.044	.906 $\pm$.003	.898 $\pm$.013	+11.35%
	Greedy	.948 $\pm$.007	.947 $\pm$.008	.589 $\pm$.045	.907 $\pm$.003	.902 $\pm$.013	+11.93%
	w/o LLM	.950 $\pm$.007	.949 $\pm$.008	.619 $\pm$.045	.906 $\pm$.003	.900 $\pm$.013	+11.67%
	TAILOR	.944 $\pm$.007	.942 $\pm$.008	.549 $\pm$.045	.908 $\pm$.003	.906 $\pm$.011	+12.40%

Remark. The Greedy and w/o LLM rows are provisional pending final measurements.

B Proofs

This appendix collects the proofs of Proposition 4.3 from Section 4 and of Proposition 5.2, Lemma 5.3, and the search bounds from Section 5.

PROOF OF PROPOSITION 4.3(I). We first record that $\succ$ is irreflexive, since $a \succ a$ would require a component to strictly exceed itself, and transitive, since componentwise $\geq$ composes and one strict gain survives, so $\succ$ admits no cycle $x_{(1)} \succ \dots \succ x_{(1)}$.

For a finite set of rows, let $\ell(x)$ be the maximum number of nodes on a domination chain $x_{(1)} \succ x_{(2)} \succ \dots \succ x_{(\ell)} = x$ ending at x (with $\ell(x) = 1$ when x has no dominator). We first show that non-dominated sorting assigns $\text{front}(x) = \ell(x)$, by induction on the front index. A point lies in front 1 iff it has no dominator, that is, iff $\ell(x) = 1$. Assume the claim holds for all fronts $< f$. A point x is placed in front f iff it is non-dominated after removing all points in fronts $< f$, that is, iff every dominator of x lies in a front $< f$ and at least one dominator lies in front $f - 1$. By the inductive hypothesis this means the longest chain of dominators ending at x has $f - 1$ strict predecessors plus x , so $\ell(x) = f$. This proves $\text{front} = \ell$.

Now suppose $a \succ b$. Take a longest domination chain ending at a , say $x_{(1)} \succ x_{(2)} \succ \dots \succ x_{(\ell(a))} = a$. Since $a \succ b$, appending b yields $x_{(1)} \succ \dots \succ a \succ b$, which is a valid chain of $\ell(a) + 1$ nodes ending at b , since validity requires only consecutive strict dominations, which hold, and the acyclicity of $\succ$ established above forbids any node from repeating. Hence $\ell(b) \geq \ell(a) + 1$, so $\text{front}(b) = \ell(b) \geq \ell(a) + 1 > \ell(a) = \text{front}(a)$. $\square$

PROOF OF PROPOSITION 4.3(II). The output gate (line 12) returns the repaired subset A when it passes both tests and A_{ref} otherwise, so there are two cases. If the gate passes, then $A^* = A$ and the two tests are exactly $G_{\text{fid}}(A^*) \geq \tau_{\text{fid}}$ and $G_{\text{trend}}(A^*) \geq \tau_{\text{trend}}$, so the floor holds. If the gate fails, then $A^* = A_{\text{ref}}$, and because $\beta \leq 1$ we have

$$G_{\text{fid}}(A_{\text{ref}}) \geq \beta G_{\text{fid}}(A_{\text{ref}}) = \tau_{\text{fid}},$$

$$G_{\text{trend}}(A_{\text{ref}}) \geq \beta G_{\text{trend}}(A_{\text{ref}}) = \tau_{\text{trend}},$$

so the reference clears the floor by construction. The derivation on line 5 is what makes the guarantee unconditional, because the thresholds are fixed as multiples of the reference’s own attained fidelity rather than taken as free external inputs, no caller can set a floor above what A_{ref} attains, and the fallback is therefore always floor-feasible. Feasibility of the floor is likewise built in, since the floor equals β times the fidelity that A_{ref} actually realizes and is therefore a level the pool demonstrably reaches. In both branches A^* meets the floor, which proves the claim unconditionally. $\square$

PROOF OF PROPOSITION 4.3(III). The candidate pool is finite and the subset size is fixed, so the set of reachable subsets is finite. Hence ϕ takes only finitely many distinct values and is bounded below by 0. Every accepted swap strictly decreases ϕ by clause (c), so the sequence $\phi(A_0) > \phi(A_1) > \dots$ of values at accepted swaps is strictly decreasing within a finite value set and therefore has finite length. Clauses (a) and (b) only shrink the set of admissible swaps and so cannot lengthen this descent, which leaves the finite-descent argument unaffected. When no accepted swap remains, no available swap simultaneously preserves the floor, holds utility, and strictly decreases ϕ , so the current subset is a local optimum with respect to the swap neighborhood. For the swap count, an accepted batch exchanges m rows and moves G_{dcr} by exactly m/k , so a non-crossing swap lowers ϕ by exactly m/k while a crossing swap is accepted only when $\phi_{\text{old}} > m/(2k)$ and leaves $\phi_{\text{new}} = m/k - \phi_{\text{old}} < m/(2k)$, after which any further move of m/k would raise ϕ and is rejected. Hence at most $\lfloor k\phi(A_0)/m \rfloor$ non-crossing swaps and at most one crossing swap are accepted, which totals at most $\lfloor k\phi(A_0)/m \rfloor$, and when $k\phi(A_0)/m$ is an integer no crossing swap can occur because the residual after the

non-crossing swaps is 0 rather than a value in $(m/(2k), m/k)$, so the ceiling holds in every case. For the terminal deviation, whenever $\phi(A) > m/(2k)$ a move of m/k toward 0.5 strictly decreases ϕ , so a swap in the improving direction satisfying (a) and (b) would be accepted, and under the stated availability assumption termination is possible only once $\phi(A_{\text{rep}}) \leq m/(2k)$, which the gate transfers to A^* on the release branch. For the floor, the input A_0 already satisfies it, since it is the output of the fidelity-constrained front-filling (line 7), and each accepted swap preserves it by clause (a), so by induction the floor is an invariant across all accepted swaps. For utility, each accepted swap satisfies $\bar{U}(A_{\text{new}}) \geq \bar{U}(A_{\text{old}})$ by clause (b), so the mean surrogate utility is non-decreasing along accepted swaps and stays at or above its pre-repair level throughout. $\square$

PROOF OF PROPOSITION 5.2. Since each $\tilde{z}_j \in [\kappa, 1]$ and $\sum_j w_j = 1$, the aggregate $\Gamma = \prod_j \tilde{z}_j^{w_j}$ satisfies $\prod_j \kappa^{w_j} \leq \Gamma \leq \prod_j 1^{w_j}$, that is, $\kappa \leq \Gamma \leq 1$. Hence $(\Gamma - \kappa)/(1 - \kappa) \in [0, 1]$ and the clip leaves it unchanged, proving $J(A) = J$.

(i) By the weighted AM-GM inequality, $\prod_j \tilde{z}_j^{w_j} \leq \sum_j w_j \tilde{z}_j$ with equality if and only if all $\tilde{z}_j$ are equal, and because $\tilde{z}_j$ is an affine bijection of z_j , this is equivalent to all z_j being equal. Applying the strictly increasing affine map $x \mapsto (x - \kappa)/(1 - \kappa)$ to both sides preserves the inequality and the equality condition, giving $J \leq J_{\text{AM}}$. Finally, since $\sum_j w_j = 1$ we have $\sum_j w_j \tilde{z}_j = (\sum_j w_j z_j + \rho)/(1 + \rho)$, whence $J_{\text{AM}} = (\sum_j w_j \tilde{z}_j - \kappa)/(1 - \kappa) = \sum_j w_j z_j$, so the baseline is exactly the plain weighted arithmetic mean of the raw components.

(ii) For each j ,

$$\frac{\partial \Gamma}{\partial z_j} = \frac{w_j}{1 + \rho} \tilde{z}_j^{w_j-1} \prod_{q \neq j} \tilde{z}_q^{w_q} > 0,$$

since $w_j > 0$ and every $\tilde{z}_q \geq \kappa > 0$. As $J = (\Gamma - \kappa)/(1 - \kappa)$ is an increasing affine function of Γ , it is strictly increasing in each z_j throughout $[0, 1]$.

(iii) Fix j and set $z_j = 0$, so $\tilde{z}_j = \kappa$ and $\Gamma = \kappa^{w_j} \prod_{q \neq j} \tilde{z}_q^{w_q}$. By part (ii), Γ is strictly increasing in each remaining z_q , so over the closed box $z_q \in [0, 1]$ it attains its maximum at $z_q = 1$ (hence $\tilde{z}_q = 1$) for all $q \neq j$, yielding $\max \Gamma = \kappa^{w_j}$ and thus $\max J = (\kappa^{w_j} - \kappa)/(1 - \kappa)$. Because $0 < \kappa < 1$ and $0 < w_j < 1$, we have $\kappa < \kappa^{w_j} < 1$, so this ceiling lies strictly in $(0, 1)$. Substituting $\rho = 0.05$ ($\kappa = 1/21$) and $w_j = 1/4$ gives ≈ 0.44 .

(iv) Part (ii) gives strict monotonicity, so moving from the dominated vector $z(A')$ to the dominating vector $z(A)$ raises at least one coordinate strictly and lowers none, whence $J(A) > J(A')$.

(v) Suppose some $A(\Theta') \in \mathcal{R}$ Pareto-dominated A^* . Part (iv) would give $J(A(\Theta')) > J(A(\Theta^*))$ and contradict the optimality of Θ^* for Eq. (8). $\square$

PROOF OF LEMMA 5.3. Since both $s_{\text{LLM}}(v_1)$ and $s_{\text{LLM}}(v_2)$ lie in $[0, 1]$ the LLM terms differ by at most 0.05 so $B(v_1) - B(v_2) \geq (R^*(v_1) - R^*(v_2)) - 0.05 > 0$. $\square$

PROOF OF THE SEARCH BOUNDS (SECTION 5.2). We first establish the value invariant by induction on the number of completed rollouts. After the first rollout the root carries its own validated reward. Each later rollout adds one validated reward R_i at a node

whose subtree membership changes exactly at its ancestors, and line 11 updates every ancestor u by $R^*(u) \leftarrow \max\{R^*(u), R_i\}$ while all other values stay untouched, so every value again equals the best validated reward in its subtree. The unconditional bound follows because each completed rollout adds one validated policy to the evaluated set and never removes one, so $\max_v R^*(v)$ is non-decreasing in the number of rollouts and always dominates the root reward $J(A(\Theta_0))$ recorded at initialization (line 1), and line 12 returns the policy attaining this maximum. The fidelity inheritance holds because every simulated subset is the output of Algorithm 1, which satisfies the floors by Proposition 4.3(ii), and the released subset is one further execution of the same operator on S under Θ^* .

For the convergence guarantee, let E_{adv} count the rounds among the first E that advance the improving sequence. Advances occur with probability at least p in every round, so E_{adv} stochastically dominates a binomial variable $\text{Bin}(E, p)$. Reaching Θ^* requires at most L advances, and the choice $E \geq \max\{2L/p, 8 \ln(1/\delta)/p\}$ gives $L \leq pE/2$, so the multiplicative Chernoff bound yields

$$\Pr[E_{\text{adv}} < L] \leq \Pr[\text{Bin}(E, p) < pE/2] \leq \exp(-pE/8) \leq \delta.$$

Once Θ^* is evaluated its reward enters the tree, and by the value invariant together with line 12 the search returns a policy of reward $J(A(\Theta^*))$, which the unconditional monotonicity preserves in every later round. $\square$

C Compressed Prompt Specification

C.1 Prompt Compression Motivation

TAILOR compresses dataset semantics, generator-source diagnosis, and rollout feedback into structured prompts. It does not provide full candidate samples, complete rollout traces, or complete experiment logs to the LLM. The LLM only proposes source-pool strategies and column-level Θ proposals, while sample-level selection is performed by deterministic post-processing and metric evaluation.

C.2 Shared Compressed Context Blocks

Table 7 lists the shared context blocks that every prompt template assembles.

C.3 Prompt Interfaces

Single-source search uses node initialization, node diagnosis, and node refine. Mixed-source search additionally uses real-data context profile summary, synthetic-source profile summary, hybrid pool initialization, and hybrid pool refine. All outputs are validated against schema, theta size bounds, target-column rules, duplicate-policy checks, and source-pool multiplier constraints. Invalid LLM outputs are repaired or replaced by deterministic fallback proposals.

D Adult Feedback-to- Θ Interpretability Case

Table 8 traces how each diagnostic signal is converted into a policy edit on *Adult*, and Table 9 lists the final searched policy components.

Table 7: Compressed prompt blocks used by TAILOR.

Block	Content	Used for
Dataset brief	Dataset name, target, column types, priority columns, target summary	Provides column semantics and task background
Theta size bounds	Min/max sizes of C_{1D} , C_{2D} , and C_P plus target rules	Constrains Θ
DCR semantics	DCR target near 0.5 and reward $1 - dcr - 0.5 $	Prevents monotonic DCR interpretation
Real utility profile	Split sizes, evaluator, metric, top utility features, semantic summary	Guides q_U
Source profile	Mean metrics, weak shape/trend, DCR imbalance, utility features	Searches mixed-source composition
Rollout feedback	Theta batch, rewards, weak columns/pairs, DCR, source contribution	Guides node diagnosis and refine
Reference nodes	Best reward node, best LLM-score node, siblings, global best	Avoids repeated proposals

Table 8: Adult feedback-to- Θ interpretability case.

Feedback signal	Observed evidence	Theta response	Interpretation
Worst shape	education.num appears among weak shape columns	Add education.num to C_{1D}	Focuses on weak marginal-distribution columns
Worst trend	marital.status-sex appears among weak trend pairs	Add marital.status and sex to C_{2D}	Converts pairwise feedback into trend constraints
Utility features	marital.status and occupation rank high	Replace or adjust q_U	Changes the utility-sensitive anchor
DCR quantiles	DCR is close to target but needs stable nearest-neighbor representation	Expand C_P	Strengthens the DCR-aware feature space

Table 9: Final Adult policy components.

Component	Role	Columns
C_{1D}	One-dimensional marginal focus	age, capital.loss, education.num, fnlwgt, income, marital.status, native.country, occupation, race, sex, workclass
C_{2D}	Two-dimensional trend anchors	age, education, education.num, fnlwgt, hours.per.week, marital.status, occupation, race, relationship, sex, workclass
C_P	DCR-aware columns	age, capital.gain, capital.loss, education, education.num, fnlwgt, hours.per.week, marital.status, native.country, occupation, race, relationship, sex, workclass
q_U	Utility balance anchor	capital.gain